

Task 1

Conversational QA System for the Story Dataset

Technical Interview Assessment - Story Dataset AI System

999

clean stories

99

actual genres

6,130

evidence chunks

50

benchmark queries

Goal. Build a chatbot that can answer questions about any story - plot, characters, settings, themes, events, summaries, metadata, and cross-story comparisons - while remaining grounded in the provided dataset.

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1. Executive summary

The final Task 1 system is a hierarchical, intent-aware RAG pipeline. Instead of treating every question as generic semantic search, a lightweight LLM planner first decides what the user is asking and which retrieval path is appropriate. Deterministic lookups are used where possible; semantic story retrieval is used for discovery; evidence chunks are retrieved for detailed QA; and canonical full stories are reserved for summarization or comparison when necessary.

This design was driven by the data. Most stories are much longer than the 256-token context used by the initial embedding model, so a single truncated embedding lost substantial information. The final retrieval layer therefore uses a story-level representation built from chunk embeddings and a separate overlapping evidence index for precise question answering.

Key outcome: retrieval quality became strong enough that most remaining answer errors were completeness/reasoning issues rather than hallucinations. The main unresolved engineering bottleneck is CPU latency, especially for long comparison prompts - a direct target for Task 3.

100% planner intent	0.95 semantic MRR	95% evidence Hit@5	100% QA grounded*
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* Groundedness is 18/18 among successfully judged QA queries; two judge calls returned no usable output.

2. Dataset understanding and quality

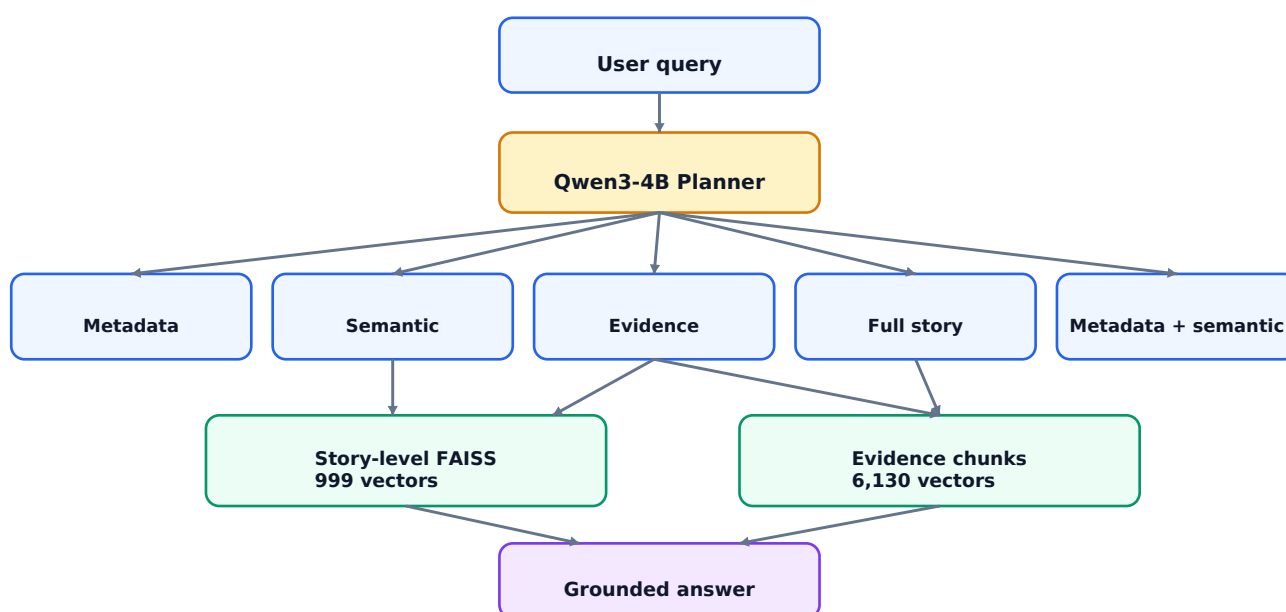
The source advertises 1,000 stories and 100 genres. Initial EDA found 1,000 rows but only 99 unique genre labels. Most genres contain 10 stories, while Historical Adventure contains 20.

Item	Finding	Engineering consequence
Raw rows	1,000	Use the full source as the starting point.
Actual genres	99	Do not hard-code the advertised 100-class assumption.
Duplicates / nulls	None found	No broad deduplication or missing-value repair required.
Malformed row	ID 308506	Excluded conservatively from the processed corpus.
Processed corpus	999 stories	Used consistently by retrieval and later classification work.
Story length	Median 978.5 words; max 3,093	Long-text handling is required.

Malformed story handling. Row 308506 ("The Chronicles of Eldareth: The Quest for the Lost Orb") contains a generation prompt rather than a real story. A conservative detector was added rather than aggressively deleting short or unusual stories. This preserves valid data while removing the verified malformed sample.

The embedding tokenizer analysis showed that about 90.2% of stories exceeded 256 tokens. This became one of the most important design signals in Task 1.

3. Final architecture



Planner: Qwen3-4B receives the user query and produces a structured plan: intent, retrieval strategy, optional titles/IDs/genre, and semantic query. Python then resolves factual identifiers; the LLM is not trusted to invent story IDs.

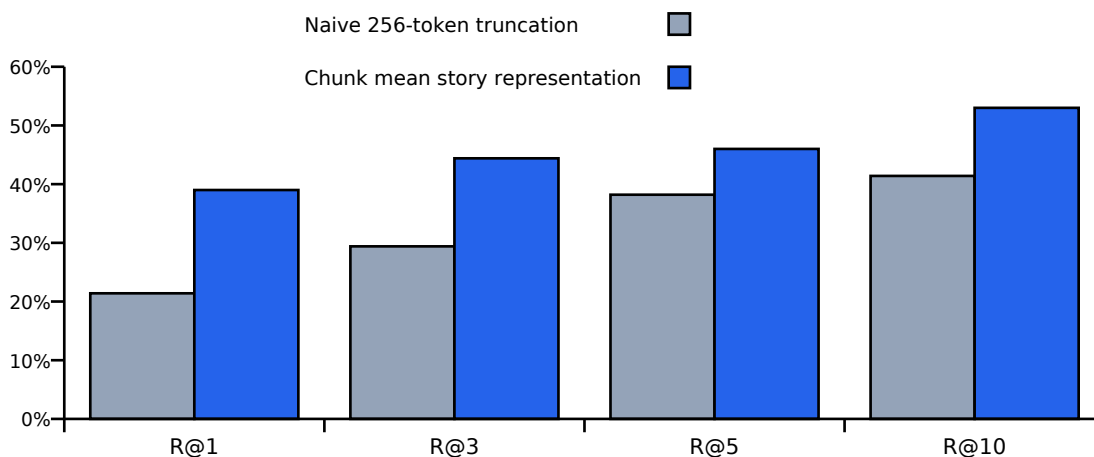
Retrieval: story-level vectors are stored in FAISS IndexFlatIP using normalized embeddings (exact cosine search). Evidence vectors are kept separately for candidate-restricted chunk search. With only 999 stories and 6,130 chunks, exact search is simpler and sufficiently fast; an approximate index such as HNSW would add unnecessary complexity.

Generation: the same local Qwen3-4B model produces context-grounded answers. Prompts are intent-specific and explicitly prohibit outside knowledge when answering story questions.

Orchestration: LangGraph routes the plan through deterministic metadata lookup, semantic discovery, evidence retrieval, full-story retrieval, or metadata-first semantic filtering, then builds context and calls the generator.

4. Retrieval journey: from naive truncation to hierarchical retrieval

The first baseline embedded only the beginning of each story. Because most stories exceed the embedding model's context window, this systematically discarded later plot events, characters, and themes.



Replacing naive truncation with non-overlapping chunk embeddings followed by mean pooling improved story retrieval across every measured Recall@K. This representation became the story-level index.

Story retrieval representation	Recall@1	Recall@3	Recall@5	Recall@10	MRR / note
First 256 tokens only	0.214	0.294	0.382	0.414	MRR 0.418
Chunk embeddings + mean	0.390	0.444	0.460	0.530	MRR 0.567

For detailed QA, the system uses overlapping evidence chunks (254-token chunks, 50-token overlap). This separates two different retrieval goals: find the right story and find the exact evidence inside that story.

5. Query planning

A 50-query benchmark was created to cover exact lookup, title lookup, metadata queries, semantic discovery, content QA, long-story QA, and cross-story comparison. The planner was evaluated separately so routing errors could be distinguished from retrieval or generation errors.

Planner metric	Result
Intent accuracy	50/50 = 100%
Retrieval-strategy accuracy	50/50 = 100%
Average CPU latency	8.60 s
P50 latency	8.66 s
P95 latency	9.67 s

Planner metric	Result
Planner errors	0

The result validates Qwen3-4B as capable enough for structured routing on the current benchmark. It also exposes a deployment problem: an ~8.6-second planner is too expensive for a low-latency CPU system. Task 3 should therefore investigate deterministic fast paths, a smaller router, caching, or prompt compression.

6. Semantic story discovery

The original benchmark used genre membership as a silver proxy for semantic relevance. Manual inspection showed this was too strict: a retrieved story could be highly relevant to a query even if its genre label differed. A separate relevance-judgment pool was therefore built, using 0 = not relevant, 1 = partially relevant, and 2 = clearly relevant.

Metric	Silver genre proxy	Content-judged relevance
Hit@1	0.20	0.90
Hit@3	0.40	1.00
Hit@5	0.60	1.00
Hit@10	0.80	1.00
MRR	0.369	0.950
Precision@5	-	0.840
Pooled Recall@5	-	0.274

Lesson: genre equality is not the same as semantic relevance. The content-judged evaluation is a better measure of the user-facing discovery task. Recall values here are pooled recall over the judged candidate pool, not exhaustive corpus recall.

7. Evidence retrieval

For content QA, the gold story ID was supplied and the system searched only that story's evidence chunks. This isolates the quality of within-story retrieval.

Subset	Hit@1	Hit@3	Hit@5	MRR
All QA queries (20)	0.50	0.80	0.95	0.660
Normal stories (15)	0.533	0.933	1.00	0.717
Long stories (5)	0.40	0.40	0.80	0.490

The main failure was Q043 in The Shadow of the Red Rose. The answer-bearing chunk existed, but ranked just beyond Top-5. This is a ranking-depth problem rather than missing information or bad chunking.

Reranker decision: a retrieve-wider + rerank stage is a justified future improvement, especially for long stories. It was intentionally deferred because dense retrieval already achieved 95% Hit@5 overall and 100% on normal stories; adding a reranker now would increase latency and complexity without a demonstrated need on most queries.

8. Answer quality and grounding

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QA answers (Q027-Q046) were judged for both correctness and groundedness using the compact retrieved evidence rather than the entire source story. This avoided unnecessarily large judge requests and measured the actual context available to the generator.

16/18 fully correct	18/18 at least partial	18/18 fully grounded	2 judge failures
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Query	Observed issue	Diagnosis
Q030 - The Crossroads of Fate	Correctly identified the US-Soviet Cold War but omitted the nuclear-war threat.	Answer completeness issue; retrieved evidence was sufficient.
Q039 - The Enigma of Elmwood Manor	Mentioned Elmwood Manor but omitted its mysterious history and Sir Reginald Blackwood.	Answer completeness issue; retrieved evidence was sufficient.
Q043 / Q046	Automated judge returned no usable result.	Excluded from the judged-rate denominator; not silently treated as correct.

The two partial answers were still fully grounded. The dominant observed failure was under-answering, not hallucination.

9. Cross-story comparison

The four comparison queries were manually adjudicated because the automated judge repeatedly returned empty outputs. Manual review used the benchmark expected answer and a 2/1/0 rubric: fully correct, partially correct, incorrect.

Query	Result	Short analysis
Q047	2/2	Correctly identified shared advanced-AI themes and consequences.
Q048	2/2	Correctly described Phantom cyberattack vs Syndicate/Oracle conflict.
Q049	1/2	Mentioned technology use in both stories but unnecessarily favored one; the expected answer was clearly 'both'.
Q050	2/2	Correctly captured both post-conflict endings.

Comparison aggregate: 3/4 fully correct (75%), 4/4 at least partially correct, 0/4 incorrect.

10. End-to-end execution and resource behavior

The first complete run used the local Qwen3-4B GGUF Q4_K_M model through llama.cpp with a 4,096-token context. The graph was built once and all 50 benchmark queries were executed.

Measurement	Result
Successful at 4,096-token context	48/50
Average latency over successful queries	28.47 s
P50 latency	25.04 s
P95 latency	46.08 s
Failures	Q047 and Q048 - comparison prompts exceeded the context window.
Q047 rerun at 8,192 context	~225.19 s

Measurement	Result
Q048 rerun at 8,192 context	~197.54 s

The 8,192-token reruns completed the answer-quality evaluation, but they are not presented as a production solution. Full-story concatenation makes comparison queries extremely slow on CPU. The production direction is evidence selection / context compression rather than simply increasing context length.

11. Model and engineering decisions

Decision	Why it was selected	Trade-off
Qwen3-4B Q4_K_M for planner + generator	Small enough for local CPU experimentation; capable of structured planning and grounded generation; GGUF/llama.cpp aligns with CPU deployment.	CPU generation is still slow; planner alone averages ~8.6 s.
all-MiniLM-L6-v2 embeddings	Compact 384-d embeddings and a fast retrieval baseline.	256-token context requires explicit long-story handling.
FAISS IndexFlatIP	Only 999 story vectors; exact normalized inner-product search is simple and fast.	Not intended as the final indexing strategy for a very large corpus.
Chunk mean for story representation	Uses information across the story and clearly beats first-256-token truncation.	Averaging can smooth away some local detail.
Overlapping evidence chunks	Improves precise within-story QA and preserves information near chunk boundaries.	Creates more vectors and retrieval work.
No reranker yet	95% overall Hit@5 and 100% on normal stories did not justify extra latency at this stage.	Long-story retrieval remains weaker.

12. Failure taxonomy

Failure type	Evidence	What it suggests
Long-story evidence ranking	Long-story Hit@5 = 80%; Q043 support was just beyond Top-5.	Retrieve wider, then rerank selectively.
Generator completeness	Q030 and Q039 omit a requested detail while remaining grounded.	Improve answer-completeness instructions or add a lightweight completeness check.
Comparison reasoning	Q049 should answer 'both', but the model favored one story.	Use a dedicated comparison prompt or structured compare-then-synthesize step.
Context budget	Q047/Q048 overflowed at 4,096 tokens.	Use evidence selection and context compression.
CPU latency	Average E2E ~28.5 s; full-story comparisons exceeded 3 minutes.	Primary optimization target for Task 3.
Automated judge reliability	Two QA and four comparison judge calls failed or returned empty output.	Use deterministic/manual fallbacks and report denominators transparently.

13. Reproducibility and project organization

The implementation separates experimentation from reusable system code. Dataset and retrieval experiments live in notebooks; reusable planner, retriever, graph, nodes, prompts, and generator live under src/; evaluation and artifact-building entrypoints live under scripts/; final indexes and metadata are stored under data/artifacts/retrieval/.

Artifact / component	Purpose
story.index	FAISS story-level exact cosine index.
story_metadata.parquet	Story ID, title, genre, and chunk count.
evidence_embeddings.npy	Dense evidence vectors for candidate-restricted search.

Artifact / component	Purpose
evidence_metadata.parquet	Chunk text, story ID, title, genre, token boundaries.
stories.parquet	Canonical processed story store.
task1_benchmark_v1.csv	50-query benchmark covering the main Task 1 behaviors.
Saved evaluation CSVs	Planner, semantic, evidence, QA, and end-to-end outputs.

14. Limitations and next improvements

1. Metadata/exact-lookup scoring. The dedicated deterministic scoring pass for Q001-Q016 was not included in the measured results summarized here. These paths are deterministic by design, but they should be scored explicitly before presenting a single consolidated benchmark table.
2. Benchmark size. Fifty queries are useful for engineering iteration but do not exhaustively cover all story facts or phrasing variations.
3. Semantic recall. Content-based Recall@K is calculated over a judged candidate pool, not over every relevant story in the corpus.
4. Long-story retrieval. Evidence quality drops on the long-story subset, supporting selective reranking or wider candidate retrieval.
5. CPU latency. The current local model proves functionality but is not yet low-latency. Task 3 should benchmark optimization choices against answer quality rather than applying techniques blindly.

Recommended Task 3 priorities: deterministic fast paths for metadata, smaller or cached planner, shorter prompts, evidence compression for comparison, thread/batch tuning in llama.cpp, and measured quantization / model-size trade-offs.

15. Conclusion

Task 1 evolved from a simple embedding idea into a measured hierarchical RAG system. The key engineering decisions were driven by observed failures: long stories invalidated naive truncation; semantic discovery required content-based relevance rather than genre-only labels; evidence retrieval separated story discovery from detailed QA; and CPU evaluation exposed context and latency limits that are invisible in accuracy-only testing.

The final system demonstrates strong routing, semantic discovery, evidence retrieval, and grounded QA on the current benchmark. Just as importantly, the remaining weaknesses are measurable and localized - long-story ranking, comparison context size, answer completeness, and CPU latency - giving Task 3 clear optimization targets.